

INFO 5602

Information Visualization

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Electives	Machine Learning Engineer (82.7 / 100)	64.4 / 100
SPECIALIZATION On-Campus Elective		

64

CURRENT AVG SCORE
across 6 job roles

96

POTENTIAL SCORE
with recommended additions

+49

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Explores the design, development and evaluation of information visualizations. Covers visual representations of data and provides hands-on experience with using and building exploratory tools and data narratives. Students create visualizations for a variety of domains and applications, working with stakeholders and their data. Covers interactive systems, user-centered and graphic design, perception, data storytelling and analysis, and insight generation. Programming knowledge is strongly encouraged.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	82.7	100.0	+17.3	MEDIUM
ML Engineer	71.1	98.5	+27.4	MEDIUM
AI Engineer	67.7	100.0	+32.3	HIGH
Data Scientist	59.5	100.0	+40.5	HIGH
MLOps Engineer	54.7	76.2	+21.5	MEDIUM
Applied Scientist	50.8	100.0	+49.2	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

LLMs & Generative AI 1 of 6 roles

storage

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1

LOW

Optional: LLMs & Generative AI — would broaden appeal to one additional role. Candidate terms: storage.
- 2

LOW

Optional: Business Communication — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

Machine Learning Engineer 82.7 → 100.0 (+17.3 pts)

interfaceslearninganalyticsolutionsbusinessmobilesciencemodels

ML Engineer 71.1 → 98.5 (+27.4 pts)

mediadiscoverylearningmachinemodelsworldproblemsnew

AI Engineer 67.7 → 100.0 (+32.3 pts)

roboticsmodelslearningsoftwaretechnicalteamsproblemstechnology

Data Scientist 59.5 → 100.0 (+40.5 pts)

toolkitanalyticsdifferentanalyticalsocialproblemsapproaches

MLOps Engineer 54.7 → 76.2 (+21.5 pts)

learningmachinemediamodels

Applied Scientist 50.8 → 100.0 (+49.2 pts)

labdiscoveryscientificcomputationalresearchstorageliteratureproblems

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.